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Transformation of *D*-xylose into furfuryl alcohol via an efficient chemobiological approach in a benign deep eutectic solvent lactic acid: betaine-water reaction system

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ABSTRACT

Biobased furfuryl alcohol (FOL) was a common building block for the production of resins, lubricants, plasticizers, organic compounds and value-added intermediates. For the effectual efficient production of FOL, a combination of chemocatalytic and biocatalytic approach was built for the transformation of *D*-xylose to FOL in a deep eutectic solvent (DES) lactic acid:betaine-H_2O (LA:Bet-H_2O) system. The furfural (FAL) yield of LA:Bet-H_2O (10:90, vol/vol) for 30 min at 180 °C reached 81.6 % using *D*-xylose (150 g/L) as substrate in a high-pressure reactor. Using glucose as co-substrate, *E. coli* HMFOMUT whole cell could transform the formed FAL (125 mM) into FOL (96.8 % yield) in a conical flask at 37 °C and pH 7.0 in this LA:Bet-H_2O (5:95, vol/vol) medium. In 50 mL LA:Bet-H_2O, HMFOMUT cells effectually convert *D*-xylose-derived FAL to FOL, and 0.75 g of *D*-xylose formed FOL (0.384 g, 78.2 % yield) and furoic acid (FA) (0.019 g, 3.4 % yield). In the present study, a one-pot tandem method was used to effectually catalyze the synthesis of FOL from *D*-xylose using an effectual combination of chemocatalysis with LA:Bet and biocatalysis with *E. coli* HMFOMUT in a reactor. It affords a new idea for the effectual use of biomass resources to synthesize high value-added compounds, and also renders a mild, green, environmentally friendly and sustainable strategy for the manufacturing of bio-based platform derivatives.

1. Introduction

The development and utilization of petrochemical resources has entirely changed the global pattern of resource supply and demand, but the excessive development and consumption of petrochemical energy has also brought hidden dangers to the ecological environment, such as ultra-high CO_2 emissions from energy combustion, which has triggered the problem of global warming [1,2]. Consequently, the exploration of sustainable and renewable resources is stimulated to replace the current unsustainable petrochemical energy economy [1–4]. As a plentiful, inexpensive, and renewable resource, biomass energy can be thermochemically processed to acquire easily degradable hemicellulose (a polymer composed of C5 and C6 monomer residues). Hemicellulose can be hydrolyzed into pentose sugars in an acidic system, with *D*-xylose being the main component of the hydrolysate [5,6]. FAL can be acquired through the dehydration of *D*-xylose [7]. FAL is a key biobased product in synthetic chemistry [8], and it is chiefly utilized in the refining of lubricating oils and in the condensation reactions with formaldehyde, phenol, acetone, or urea to produce resins with excellent thermal stability and high physical strength [9,10].

FAL can be utilized as a starting material to manufacture a series of valuable furan derivatives, such as furoic acid (FA), furfuryl alcohol (FOL) and furfurylamine (FLA), through the oxidation, hydrogenation, amination, and other reactions [2,8,9,11]. Acidic catalysts can be mainly used to dehydrate *D*-xylose into FAL [8–11]. The catalyst pTSA-$CrCl_3{\cdot}6H_2O$ transformed *D*-xylose to FAL in 53 % yield at 120 °C [12]. *D*-Xylose was dehydrated to FAL (72 % yield) at 140 °C with sulfonated graphene [13]. However, traditional hydrolysis processes require high loads of inorganic acids, which can result in severe equipment corrosion and environmental pollution [8]. Different from traditional organic solvents and ionic liquids (IL), deep eutectic solvent (DES) has the advantages of low production cost, convenient synthesis, easy adjustment and non-toxic [14]. DES is a uniform transparent liquid formed by mixing and stirring hydrogen bond acceptor (HBA) and hydrogen bond

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donor (HBD) at a certain temperature [8,15]. DES-H_2O has been applied to catalyze the conversion of biomass (such as bread waste) and biomass derivatives (such as sugars and xylose) into furans [16]. In a DES GA: ChCl medium, the resin Amberlyst 15 can promote the dehydration of *D*-fructose to produce 5-hydroxymethylfurfural (78 % yield) [17]. In recent years, DESs have received extensive attention as a reaction solvent and/or chemical catalyst for FAL production. The DES (CA: Betaine)–H_2O catalyzes the conversion of carbohydrates in corn cobs (60 g/L) into FAL (74.6 mM), formic acid (96.5 mM), and glucose (40.3 mM) at 170 °C for 30 min [11]. Using DES ChCl/MA/LA (15 wt%) as chemocatalyst, *D*-xylose was dehydrated to acquire FAL at 190 °C for 45 min, with a yield of 70.6 % [2]. Under the presence of formic acid in DES ChCl:Gly (7.5 vol%), biomass-derived *D*-xylose can be transformed at 180 °C for 30 min, resulting in the generation of FAL with a yield of 58.3 % [8].

Recently, it is of great interest to synthesize valuable furan-based chemicals via biocatalysis [8,9,11,19–22]. FOL as one of the derivatives of FAL is extensively used in the production of vitamin C, lysine, plasticizers, adhesives, cement, fiberglass, furan resins, lubricants, polymer concrete and other products [18–21]. Accordingly, attempts have been made to catalyze the biosynthesis of FOL from FAL by bioreduction. Different from the traditional chemical reduction method, the bioreduction method is environmentally friendly and has mild performance conditions. The FOL was synthesized by *E. coli* CG-19 catalyzed by 150 mM FAL at pH 7.5 and 35 °C for 24 h, acquiring the FOL yield of 75.1 % [19]. *Bacillus coagulans* NL01 catalyzed 52 mM FAL to acquire FOL. The FOL yield was 73.0 % at pH 7.0 and 50 °C for 24 h [20]. However, biosynthesis of FOL from FAL in single water system suffers from long reaction time, low substrate loading and low selectivity. DES as a good biocompatible catalytic solvent, can lead to the increased enzyme stability and positively affect the selectivity of the reaction when DES-H_2O was used as the reaction medium [21,23]. Recently, DESs based on betaine have manifested good biocompatibility, and carboxylic acid-based DES can be used for the dehydration of *D*-fructose to obtain BHMF [21]. Acidic DESs, known for their biocompatibility, can serve as both chemocatalysts and reaction media for the dehydration of *D*-xylose to produce FAL [8]. Consequently, the establishment of a suitable DES system contributes to the effectual biocatalysis [23]. The DES-mediated conversion of *D*-xylose to FOL represents a sustainable pathway. To date, there have been relatively few studies on the catalytic synthesis of FOL from *D*-xylose through tandem chemical and biocatalytic reactions in DES systems. One-pot chemobiocatalyitc cascade strategy for effectually transforming *D*-xylose into FOL can be attempted via chemocatalysis with DES and biocatalysis with reductase biocatalyst.

Recombinant *E. coli* HMFOMUT cell containing double mutant HMFO oxidase, which was obtained from the mutation of HMFO in *Methylovorus* sp. Strain MP688 (valine367 was mutated to arginine, and tryptophan 466 was mutated to phenylalanine) [22], was applied to transform *D*-xylose-derived FAL into FOL. To efficiently convert *D*-xylose to FOL, one chemobiological catalysis strategy was built via a sequential conversion of *D*-xylose with DES lactate:betaine (LA:Bet) and HMFOMUT cell in one-pot manner. This study applied LA:Bet as a chemocatalyst to prepare FAL from *D*-xylose in a DES-H_2O system, and *D*-xylose-derived FAL was biologically converted into FOL by HMFOMUT cells harboring reductase in LA:Bet-water system. The effect of different ratios of LA:Bet-H_2O system on the chemocatalytic synthesis of FAL from *D*-xylose was testified. Factors such as system pH, bioreduction temperature, co-substrates and precursors, metal ions, and FAL concentration in the DES-H_2O system were examined in relation to their effects on the enzymatic conversion of FAL to FOL by HMFOMUT cells. Additionally, one-pot chemobiocatalyitc cascade strategy for effectually transforming *D*-xylose into FOL was testified via chemocatalysis with LA:Bet and biocatalysis with HMFOMUT cell. By integrating chemical and biological approaches in an eco-friendly reaction medium, a new pathway for the effectual synthesis of FOL from *D*-xylose was constructed through the one-pot tandem chemobiological catalysis.

2. Materials and methods

2.1. Materials

Furfural (FAL), furfuryl alcohol (FOL), furoic acid (FA), *D*-xylose, lactic acid (LA), and betaine (Bet) reagents were from Lingfeng Chemical Reagents Co., Ltd. (P.R. China).

2.2. Synthesis of DES

In this research, a DES lactate:betaine (LA:Bet) was prepared. Lactic acid served as the hydrogen donor, and betaine acted as the hydrogen acceptor. In a 300 mL round-bottomed flask, the mixture was made according to molar ratios of 1:1, 1.5:1, 2:1 and 3:1, respectively. The mixture was incubated at 65 °C with magnetic stirring at 400 rpm for 1 h until a homogeneous and transparent solution was acquired. The resulting LA:Bet DES was used for both chemocatalytic and biocatalytic reactions.

2.3. Synthesis of FAL through the dehydration of D-xylose in DES-H_2O

A mixture of 1.5 g of *D*-xylose and an acidic DES catalyst (0–30 wt%) was thoroughly mixed in 50 mL of deionized water. The mixture was then transferred to a 100 mL high-pressure reactor (TGYF-B 100 mL QIYAO, Shanghai) and stirred at 170–190 °C and 500 rpm for 8–120 min to dehydrate *D*-xylose into produce FAL. The samples were analyzed using high-performance liquid chromatography (HPLC). The FAL yield was calculated as follows:

$$\text{FAL yield } (\%) = \frac{\text{FAL produced}(mM)}{\text{Initial D - xylose}(mM)} \times 100 \quad (1)$$

2.4. Determination of Kamlet-Taft parameters

The Kamlet-Taft parameters (α, β and π^*) were acquired according to the previous reports [24]. Where *N,N*-dimethyl-4-nitroaniline is applied as the indicator for π^*, Nile red dye and 4-nitroaniline are the indicators for α and β, respectively [25]. λ_{max} denotes the maximum value of the wavelength. ν, α, β and π^* values were calculated according to the following equations:

$$\pi^* = 0.314 \times (27.52 - \nu_{NEt2}) \quad (2)$$

$$\beta = 11.134 - \frac{3580}{\lambda(NH_2)_{max}} - 1.125\pi^* \quad (3)$$

$$\alpha = \frac{19.9657 - 1.0241\pi^* - \nu_{NR}}{1.6078} \quad (4)$$

$$\nu = 1/(\lambda_{max} \times 10^{-4}) \quad (5)$$

2.5. Biosynthesis of FOL through the bioreduction of FAL in DES-H_2O

The gene fragment of the HMFO dehydrogenase from the strain *Methylovorus* sp. MP688 was synthesized by Jinkairui (Wuhan, China) [22]. The amino acid leucine at position 367 was mutated to arginine, and tryptophan at position 466 was mutated to phenylalanine, acquiring double mutant (HMFOMUT). The constructed *E. coli* HMFOMUT carrying this double mutant was constructed and inoculated on Luria-bertani (LB) agar plates containing antibiotic (50 μg/mL kanamycin). Single colony was inoculated into a glass tube containing 3.0 mL of LB medium containing antibiotic (50 μg/mL kanamycin) and incubated overnight at 37 °C. 500 μL of pre-inoculum was supplemented to the Terrific Broth (TB) medium and incubated on a shaker (37 °C, 220 rpm) until the OD_{600} of the cells reached 0.6–0.8. Afterwards, isopropyl β-*D*-

thiogalactopyranoside (IPTG) as inducer was supplemented to TB medium, reaching a final concentration of 0.2 mM. After HMFOMUT cells were induced (22 °C) for 15 h, cells were collected by centrifugation (4000*g*, 10 min, 4 °C). The cell sediment was washed twice with phosphate buffer (100 mM, pH 7.0), and whole cells of HMFOMUT were collected for biocatalysis.

The catalytic system and reaction conditions for HMFOMUT catalyzed synthesis of FOL from commercial FAL were optimized. The effect of DES LA:Bet dose (0–30 wt%) on the bioreduction of FAL into FOL was testified. Afterwards, the effects of temperature (20–60 °C), pH (6.0–10.0), metal ions (Fe^{3+}, Cr^{3+}, Ba^{2+}, Co^{2+}, Ni^{2+}, Ca^{2+}, Sn^{4+}, Zn^{2+}, Mg^{2+}, Fe^{2+}, Mn^{2+} and Al^{3+}) (1 mM), co-substrate and precursor concentrations on the whole-cell biocatalytic activity of HMFOMUT were assessed. HMFOMUT (D_{600} = 40.0) and FAL (75 mM) were added to a 25 mL conical flask in a 5 mL system, and after incubation on a shaker (200 rpm) for a certain reaction time, FAL, FOL, and FA were determined by HPLC. Their yields were calculated as follows:

$$\text{FOL yield}(\%) = \frac{\text{FOL produced(mM)}}{\text{Initial FAL (mM)}} \times 100 \quad (6)$$

$$\text{FA yield}(\%) = \frac{\text{FA produced (mM)}}{\text{Initial FAL (mM)}} \times 100 \quad (7)$$

$$\text{FOL Selectivity}(\%) = \frac{\text{FOL produced(mM)}}{(\text{FA + FOL}) \text{ produced (mM)}} \times 100 \quad (8)$$

2.6. HPLC analysis

Through the analysis by HPLC (LC-2030C 3D SHIMADZU, Japan) containing a Discovery® C18 (3.9 mm × 150 mm, 4 μm) HPLC column, FAL, FOL and FA were eluted with a mobile phase consisting of 80 vol% water, 20 vol% methanol and 0.1 wt% of trifluoroacetic acid (TFA) at a column temperature of 35 °C. The flow rate was 0.80 mL per minute. FAL and FA were detected at a wavelength of 254 nm. FOL was monitored at a wavelength of 210 nm.

3. Results and discussions

3.1. Enhanced production of FAL from D-xylose through the dehydration with LA:Bet-H_2O

FAL is a kind of multifunctional chemical platform molecule, which can be acquired through the dehydration of *D*-xylose with acid catalyst (Fig. S1, See Supplementary material) [26]. The traditional hydrolysis process requires a homogeneous catalyst represented by a high load of inorganic acid, which will lead to low conversion, selective, serious equipment corrosion and other drawbacks [8,27]. Consequently, it is very important to apply suitable acidic catalyst in the dehydration process of *D*-xylose. In recent years, DES, as a new sustainable solvent, has the characteristics of simple synthesis process, high stability and low cytotoxicity, and can effectually degrade biomass and acquire high value-added compounds [28–31]. In the field of green bioenergy has been extensively concerned.

In this research, the effects of dehydration time, dehydration temperature, LA:Bet dose and *D*-xylose dosage on the dehydration of *D*-xylose into FAL were testified in DES-H_2O system. Kamlet-Taft parameter is an important physical and chemical property, which is a key factor reflecting the overall solvation ability [32–35]. In the process of chemical reaction, it will affect the reaction rate, product yield, etc [32]. In DES, α, β and π^* represent the ability to provide H^+, the ability to receive H^+ and the polarity/polarizability, respectively [33]. When the α value of DES is higher, the H^+ release ability is stronger, and when the β value of DES is higher, the ability to accept H^+ is stronger [35]. Therefore, α and β are critical to the efficiency of pretreatment of biomass and its derivatives. As showcased in Fig. 1, when the molar ratio of LA:Bet was 3:1, 2:1 and 1.5:1, respectively, all remained transparent after cooling by fully stirring. When the molar ratio of LA:Bet was 1:1, the mixture was semi-solid, indicating that the mixture was not uniform. With the gradual increase of the proportion of HBD to HBA in DES LA:Bet, α value increased from 1.30 to 1.38, β value decreased from 0.57 to 0.45, and π^* value declined from 1.07 to 1.03 (Table 1). The synthesis of FAL from *D*-xylose was catalyzed by 15 wt% DES LA:Bet as chemocatalyst at 180 °C for 30 min with HBD to HBA molar ratios of 1.5:1, 2:1, and 3:1, respectively. The results showcased that the FAL yield was higher when the molar ratio of HBD to HBA of LA:Bet was 2:1 (yield 74.5 %) and 3:1 (yield 63.7 %) than that of 1.5:1 (yield 54.0 %) (Fig. 2). When the HBD to HBA molar ratios were 3:1 and 2:1, they had higher α-values and lower β-values compared to HBD to HBA molar ratios of 1.5:1, which made it easier for H^+ bonds to be generated, and thus facilitated the hydrolysis efficiency of DES [25,35].

The catalytic temperature, time and amount of DES will affect the synthesis of *D*-xylose by DES [26]. As showcased in Fig. 3a, when the dehydration temperature rose from 170 °C to 180 °C, the FAL yield from *D*-xylose was elevated from 69.1 % to 74.5 % at 30 min in the DES LA:Bet-H_2O system. As the temperature further elevated to 190 °C, the yield

Table 1
Kamlet–Taft parameters and viscosity for different deep eutectic solvents.

Experiment No.	DES	HBD/HBA molar ratio	Kamlet-Taft parameters		
			π^*	α	β
1	LA:Bet	1:1	/	/	/
2	LA:Bet	1.5:1	1.07	1.30	0.57
3	LA:Bet	2:1	1.06	1.33	0.52
4	LA:Bet	3:1	1.03	1.38	0.45

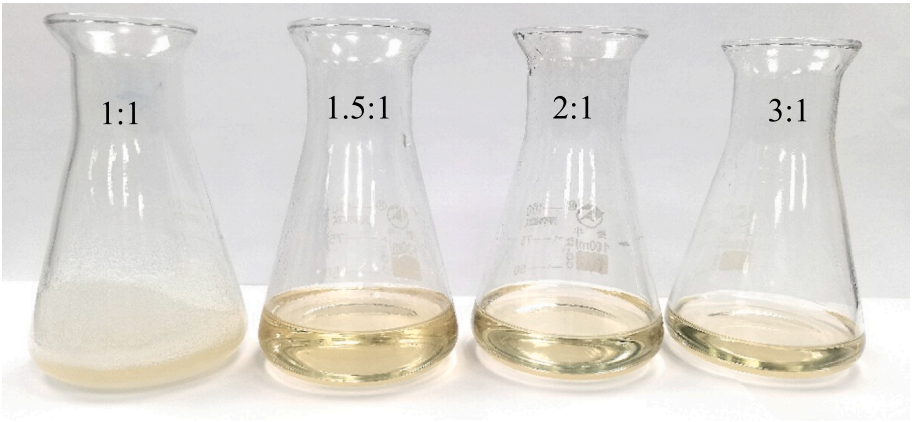


Fig. 1. DES LA:Bet with different HBD-to-HBA molar ratios (1:1, 1.5:1, 2:1 and 3:1).

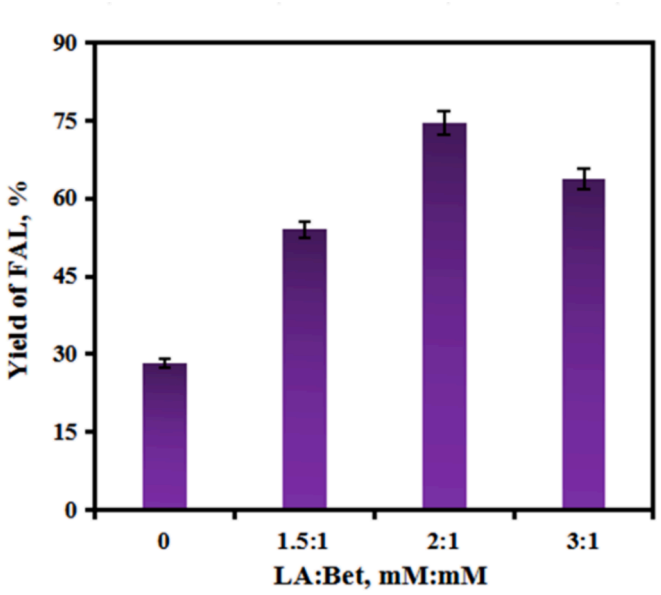


Fig. 2. Effect of different ratios of HBD-to-HBA in DES LA:Bet (1.5:1, 2:1 and 3:1) on the catalytic synthesis of FAL from *D*-xylose [LA:Bet 15 wt%, 180 °C, 30 min].

of FAL declined slightly, reaching 70.3 % yield. This might be due to a side-reaction of rehydration or condensation occurring at higher reaction temperatures [36]. Accordingly, the optimal dehydration temperature was 180 °C. At 180 °C, the concentration of FAL varied at different synthesis time (8 min-120 min). As the reaction duration was rose from 8 min to 30 min, the FAL yield augmented from 66.9 % to 74.5 %. When the reaction duration was raised from 30 min to 90 min, the FAL yield was weakened from 74.5 % to 61.7 %. As the reaction duration increased, FAL might be further degraded to insoluble and soluble black substances, such as coke, thereby weakening the concentration of FAL [37]. The effect of DES LA:Bet dose on the dehydration of *D*-xylose to FAL was testified at 180 °C for 30 min. Upon adding DES LA:Bet from 0 wt% to 10 wt%, the FAL yield increased substantially from 28.4 % to 74.5 %. Excess water molecules near the active catalytic site would produce competitive hydrolysis reactions, thus forming a diffusion barrier for hydrophobic substances [38]. In the system of DES LA:Bet-H_2O, the appropriate addition of DES LA:Bet would weaken the competitive hydrolysis of excessive water molecules at the active site, thus promoting the stability of the chemical catalytic process. By raising DES LA:Bet loading from 10 wt% to 30 wt%, the FAL yield dropped from 74.5 % to 64.3 %. The increase of catalyst addition might influence the mass and heat transfer of the reactants and products in and between phases, thus influencing the performance of the reaction system [39,40]. At 180 °C for 30 min, the optimal dose of DES LA:Bet was selected as 10 wt% (Fig. 3b). The effect of *D*-xylose dose on FAL yield was testified in DES-H_2O system (10 wt% LA:Bet) catalyzed at 180 °C for 30 min. In the 50 mL DES-water system, the addition of *D*-xylose evevated from 100 mM to 800 mM, and the FAL yield declined from 81.6 % to 62.1 % (Fig. 3c). This was due to the insufficient contact between the excessive substrates and chemocatalysts in a limited reactor [21,41,42]. As a chemocatalyst, Nb_2O_5 catalyzed *D*-xylose to synthesize FAL at 120 °C and 180 min, with a yield of 48 % [43]. SO_4^{2-}/SnO_2-APG was used as a catalyst to synthesize FAL from *D*-xylose under 170 °C for 20 min, with a yield of 41.9 % [44]. Compared with Nb_2O_5 and SO_4^{2-}/SnO_2-APG as catalysts, DES LA:Bet catalyzed *D*-xylose dehydration to synthesize FAL more effectually in a shorter period of time.

In general, DES LA:Bet as a catalyst has better catalytic effect than Nb_2O_5, SO_4^{2-}/SnO_2-APG ChCl/MA/LA as a catalyst for the dehydration of *D*-xylose to synthesize FAL (Table 2). 0.75 g *D*-xylose was dehydrated to FAL in DES LA:Bet-H_2O (10:90 wt/wt) system at 180 °C for 30 min,

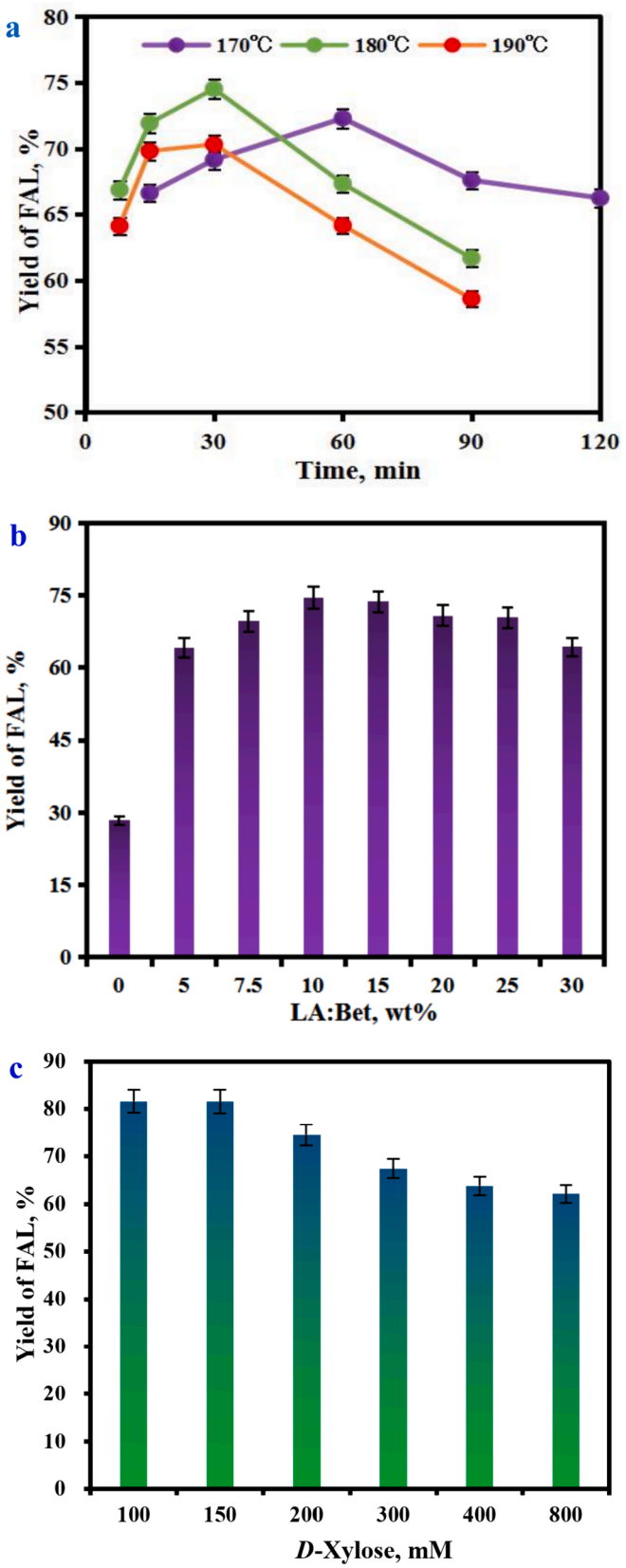


Fig. 3. Effects of time (8–120 min) and temperature (170–190 °C) in LA:Bet-H_2O on FAL generation [200 mM of *D*-xylose, DES LA:Bet 15 wt%] (a); Effects of DES LA:Bet load (0–30 wt%) in DES-H_2O on FAL production [200 mM of *D*-xylose, 180 °C, 30 min] (b); Effects of *D*-xylose load (100 – 800 mM) on FAL formation [LA:Bet 10 wt%, 180 °C, 30 min] (c).

Table 2
The formation of furfural catalyzed by different catalysts from *D*-xylose.

Catalyst and dosage	Solvent	Operation condition	Yield of FAL, %	Ref.
Nb_2O_5 (0.1 g)	H_2O	120 °C, 180 min	48	[43]
SO_4^{2-}/SnO_2. APG (2.4 g)	H_2O	170 °C, 20 min	41.9	[44]
BE:Bet (0.8 g)	H_2O	170 °C, 30 min	51.1	[26]
ChCl:Gly (2.5 g)	H_2O	170 °C, 30 min	58.3	[8]
ChCl/MA/LA (5 g)	H_2O	190 °C, 45 min	70.6	[2]
LA:Bet (5 g)	LA:Bet-H_2O	180 °C, 30 min	81.6	This research

and the FAL molar yield reached 81.6 %.

3.2. Biosynthesis of FOL from FAL through the bioreduction with HMFOMUT in DES LA:Bet-H_2O

In recent years, biocatalysis of FAL to FOL has attracted the attention of researchers due to its merits such as low energy consumption, high selectivity, mild bioreaction conditions and high selectivity of FOL [45,46]. Using glucose as a cosubstrate, CCZU-Y10 cells could biologically reduce FAL into FOL, and the yield of FOL reached 44 % within 150 min [47]. The method of biocatalysis was recognized as an effectual alternative to chemical catalysis. In the process of biological reaction, the bioreaction system has a crucial role in the stability of biocatalysis. Water is currently the most extensively utilized reaction solvent in biocatalytic reactions, and controlling water content is crucial for some biocatalytic reactions [48,49]. When DES and water are mixed as bioreaction media, the enzyme can be more stable, the biocatalytic efficiency of the enzyme can be augmented, and the selectivity of the reaction can be positively affected [25,50]. Accordingly, the effect of LA:Bet loading in DES-H_2O system (LA:Bet 0–30 wt%) on whole-cell catalytic synthesis of FOL from 75 mM FAL by HMFOMUT was testified. As the addition level of LA:Bet was lower than 5 wt%, increasing the addition level of LA:Bet could enhance the biocatalytic activity of whole-cell conversion of FAL to FOL with HMFOMUT cells. When the addition of LA:Bet was 5 wt%, the whole cells of HMFOMUT manifested the best biocatalytic activity, and the relative activity of FOL was 2.9-times higher than that in single water system, and the selectivity of FOL also reached 98.6 %. The catalytic activity of HMFOMUT cells gradually declined when LA:Bet was added above 7.5 wt%. The catalytic activity remained high compared to that without the addition of LA:Bet. As the dose of LA:Bet was 25 wt%, the catalytic activity of HMFOMUT whole cells was still high, and its catalytic activity was increased by 1.8-times. Compared with LA:Bet (5 wt%), the catalytic activity of HMFOMUT whole cells weakened. When the dose of LA:Bet was 30 wt%, the catalytic activity of HMFOMUT cells was only augmented by 1.2-times. At 37 °C and pH 7.0, the biocatalytic activity in the existence of 2.5–30 wt% LA:Bet was 1.2–2.9 times higher than that without LA:Bet. This might be due to that LA:Bet could evelate the whole cell reaction rate and stability of HMFOMUT [49]. Accordingly, the optimal addition of LA:Bet was 5 wt% (Fig. 4a). The pH of the reaction medium is known to play a key role in biotransformation. Buffer pH may alter the ionic state of reductases and substrates involved in the reduction process [2]. The influences of different pH values (5.0–10.0) on whole-cell catalysis of FAL to acquire FOL by HMFOMUT in LA:Bet-H_2O (5:95, wt/wt) system at 37 °C were testified. The results indicated that the catalytic activity of HMFOMUT cells elevated gradually as pH increased from 5.0 to 7.0. When pH was higher than 7.0, the catalytic activity of HMFOMUT cells declined gradually. This might be due to that over-acid or over-base reaction conditions could result in enzyme inactivation, thereby weakening catalytic activity [16,21]. When pH was 7.0, it manifested the optimal biocatalytic activity (Fig. 4b). The biological reaction temperature would affect the activity and stability of biocatalysis and the balance of the reaction [8,51]. Consequently, the effect of different temperatures on whole-cell catalytic FAL synthesis of FOL by HMFOMUT in LA:Bet-H_2O (5:95, wt/wt) system at pH 7.0 was testified. The results manifested that the whole cell biocatalytic activity of HMFOMUT elevated substantially when the temperature increased from 20 °C to 37 °C. Over 37 °C, the whole-cells activity of HMFOMUT declined. As the reaction temperature was 37 °C, the whole cells of HMFOMUT manifested the optimal catalytic activity (Fig. 4c). Accordingly, in LA:Bet-H_2O systems, the optimal pH and temperature were 7.0 and 37 °C, respectively.

Because dehydrogenase has the main characteristic of metal ion dependence [52]. Accordingly, the effects of different metal ions (Fe^{3+}, Cr^{3+}, Ba^{2+}, Co^{2+}, Ni^{2+}, Ca^{2+}, Sn^{4+}, Zn^{2+}, Mg^{2+}, Fe^{2+}, Mn^{2+} and Al^{3+}) (1 mM) on whole-cells bioreduction of FAL into FOL were testified by HMFOMUT cells in LA:Bet-H_2O system. The results showcased that Fe^{3+} had no inhibitory effect on the whole cell catalytic activity of HMFOMUT, while other metal ions had inhibitory effect on the cells catalytic activity. Al^{3+} weakened the biocatalytic activity of HMFOMUT to 28.3 %. The inhibition effect on the bioreduction might be due to that Al^{3+} might occupy the binding site and inhibit biocatalytic activity [19]. Accordingly, these metal ions didn't manifest a catalytic effect on the whole cell biocatalysis of HMFOMUT, as showcased in Fig. 4d. Cosubstrates and precursors play a key role in the biological reduction process [53]. Studies have manifested that cosubstrate glucose might substantially improve the catalytic activity of FAL reduction [52,53]. Amino acid supplementation can enhance detoxification of microbial cells, thereby improving oxidation–reduction catalytic reactions [53] The effects of different cosubstrates and precursors (100 mM) on whole-cells catalysis of FAL to FOL by HMFOMUT in LA:Bet-water (5:95, wt/wt) at pH 7.0 and 37 °C were testified. It was noticed that glucose, *D*-fructose and *D*-xylose might promote the transformation of FAL. Compared with other cosubstrates, glucose showed more substantial promoting effect. However, amino acid precursors *D*-histidine and *L*-lysine exhibited inhibition of FOL generation (Fig. 4e). This might be due to the fact that these amino acids inhibit the synthesis of dehydrogenase cofactors [54], thereby inhibiting the whole-cell bioreduction of FAL into FOL with HMFOMUT. The amount of cosubstrate added might have a certain impact on the bioreaction system [44]. Accordingly, in a LA:Bet-H_2O (5:95, wt/wt) system at pH 7.0 and 37 °C, the effect of glucose dose on whole-cell catalytic synthesis of FOL by 75 mM HMFOMUT was further testified. When the molar ratio of glucose to FAL evelated from 0:1 to 0.8:1, the biocatalytic activity rose in gradual manner. In the existence of cosubstrate glucose (0.8 mol glucose/mol FAL), the optimal catalytic activity of HMFOMUT whole cells was acquired, and the catalytic activity was 1.2-times higher than that without cosubstrate. When the molar ratio exceeded 0.8:1, the biocatalytic activity of whole cells of HMFOMUT declined. Biocatalysis was inhibited when the molar ratio exceeded 5:1 (Fig. 4f). This might be ascribed to that the viscosity of the reaction medium might augment with the increase of the cosubstrate, thereby weakening the activity of dehydrogenase [21,44].

The titer of substrate can substantially affect the catalytic activity of cells [21]. The effects of FAL at different concentrations (50–200 mM) on whole-cell synthesis of FOL from HMFOMUT were measured with cosubstrate glucose (0.8 mol glucose/mol FAL) in LA:Bet-H_2O (5:95, wt/wt) at pH 7.0 and 37 °C. When FAL was $\leq$ 125 mM, a entire biotransformation of FAL into FOL was acquired within 24 h. 50 mM and 100 mM of FAL could be entirely converted within 8 h, and the selectivities of FOL were 95.2 % and 97.8 %, respectively. When the FAL concentration was 125 mM, the FAL conversion rate and selectivity were 97.4 % and 97.0 % at 8 h, respectively. As the FAL concentration exceeded 125 mM, FAL was not fully converted. Using 150 mM and 200 mM FAL as substrates, the conversion rates of FAL in 24 h were 80.7 % and 40.1 %, respectively. The selectivity of FOL was over 92 %. With the elevated FAL concentration, the catalytic activity of HMFOMUT whole cells gradually declined, as showcased in Fig. 5. The FOL was acquired through the bioreduction of 150 mM FAL with *E. coli* CG-19 at pH 7.5 and 35 °C for 24 h, and the FOL yield was 75.1 % [19]. *Bacillus coagulans*

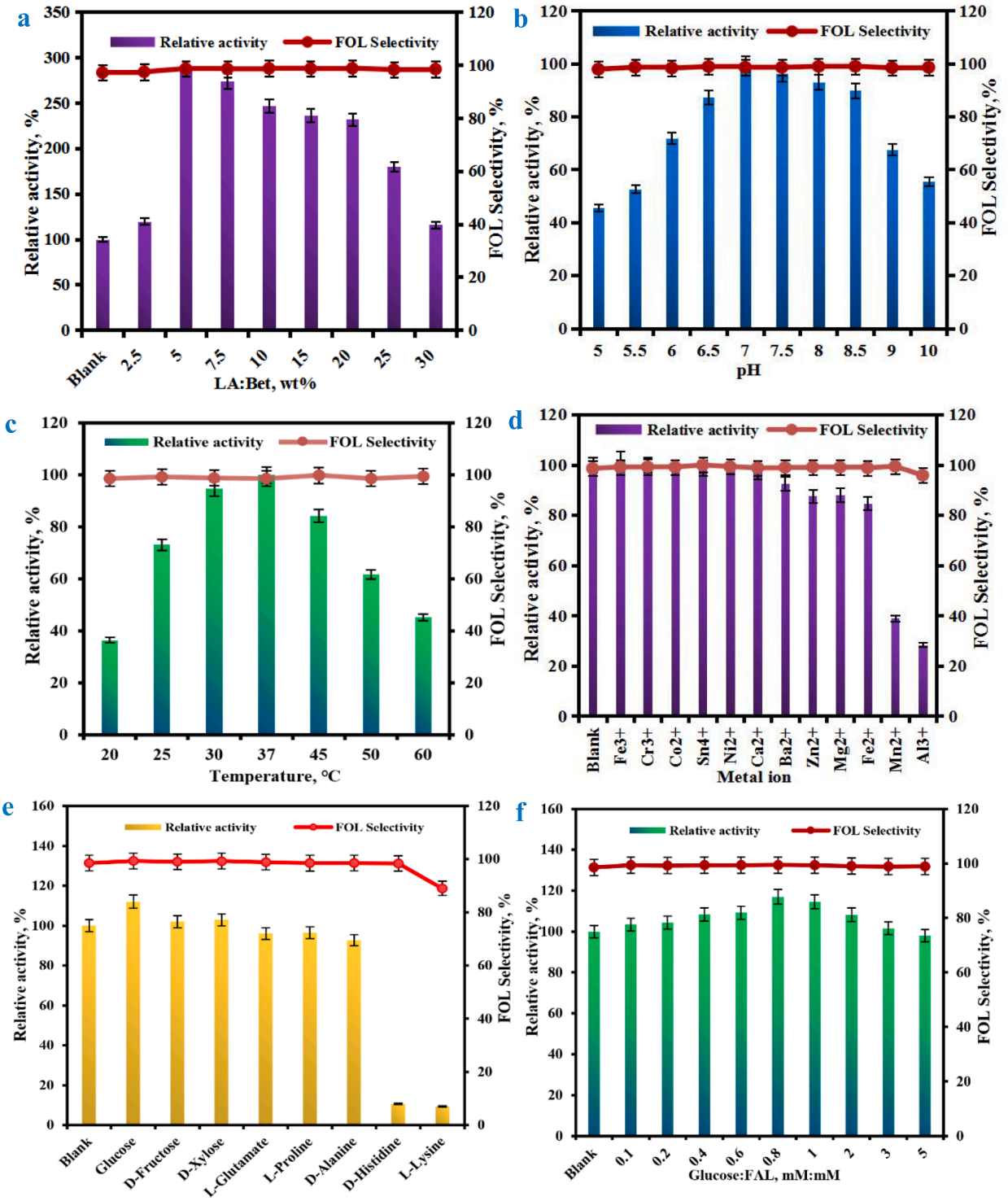


Fig. 4. Effects of LA:Bet load (0–30 wt%) on the bioreduction [75 mM FAL, 37 °C, pH 7.0] (a); Effects of medium pH (5.0–10.0) on the bioreduction [LA:Bet 5 wt%, 75 mM FAL, 37 °C] (b); Effects of bioreaction temperature on the bioreduction at 20–60 °C [LA:Bet 5 wt%, 75 mM FAL, pH 7.0] (c); Effects of additive (1.0 mM) on the bioreduction [LA:Bet 5 wt%, 75 mM FAL, 37 °C, pH 7.0] (d); Effects of co-substrates and precursors (100 mM) on the bioreduction [LA:Bet 5 wt%, 75 mM FAL, 37 °C, pH 7.0] (e); Effects of co-substrate glucose load on the bioreduction [LA:Bet 5 wt%, 75 mM FAL, 37 °C, pH 7.0] (f).

NL01 catalyzed 52 mM FAL at pH 7.0 and 50 °C for 24 h to synthesize FOL, acquiring a yield of 73.0 % [20]. *Bacillus coagulans* NL01 catalyzed 208 mM FAL to synthesize FOL. The FOL yield was 88.0 % after reaction at pH 7.0 and 50 °C for 24 h [55]. Whole cells of HMFOMUT could effectually convert 50–150 mM FAL to FOL within 24 h, with FOL yield ranging from 80.7 % to 97.0 %.

In a concise summary, the optimal conditions for catalyzing the synthesis of FOL from FAL using HMFOMUT whole-cells catalyst were as follows: LA:Bet 5 wt%, pH 7.0, 37 °C, with a molar ratio of glucose to FAL as 0.8:1. Distinct from other biocatalytic cells such as *Bacillus coagulans* NL01, *E. coli* CCZU-Y10, *E. coli* CG-19, and *E. coli* CF, HMFOMUT whole-cells demonstrates effectual conversion of FAL to FOL under milder reaction conditions and with fewer co-substrates (Table 3) [8,19,20,47,55].

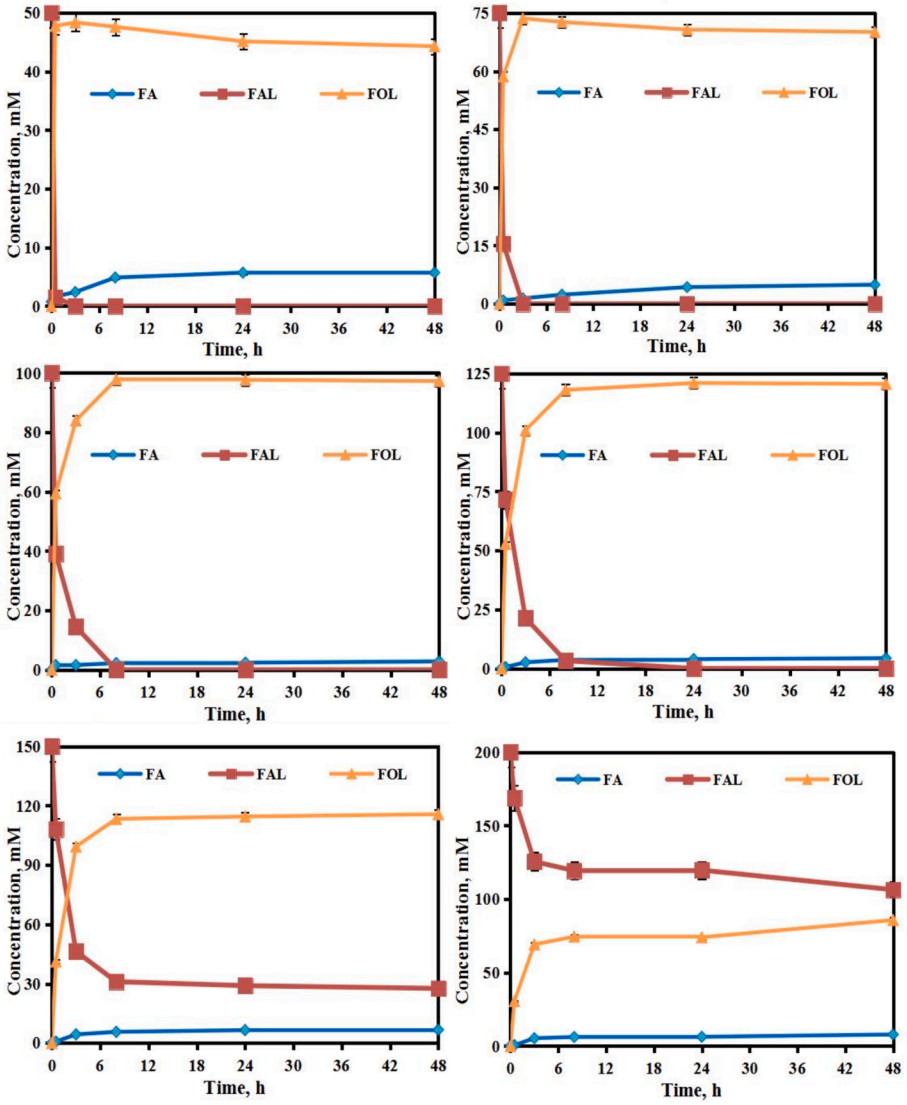


Fig. 5. Biological conversion of FAL (50–200 mM) with HMFOMUT cells and glucose (FAL-to-glucose molar ratio, 1:0.8) at 37 °C and pH 7.0 in LA:Bet-water (5 wt %) system.

Table 3
The synthesis of FOL from FAL catalyzed by different biocatalysts.

Biocatalyst	Biocatalyst reaction conditions	Cosubstrate and dosage	Yield of FOL	Ref.
Bacillus coagulans NL01	pH 7.0, 50 °C, 52 mM FAL, 24 h	glucose 55.5 mM	73.0	[20]
E. Coli CCZU-Y10	pH 6.5, 30 °C, 100 mM FAL, 24 h	glucose 14.5 mM xylose 2.4 mM Arabinose 2.1 mM	88.6	[47]
E. coli CG-19	pH 7.5, 35 °C, 150 mM FAL, 24 h	SDS 0.50 mM, glucose 225 mM $NADP^+$ 0.15 mM	75.1	[19]
E. coli CF	pH 6.5, 35 °C, 100 mM FAL, 24 h	ChCl:Gly, 7.5 vol%, HCOOH 3.0 wt%	96.6	[8]
Bacillus coagulans NL01	pH 7.0, 50 °C, 208 mM FAL, 24 h	glucose 222.0 mM	88.0	[55]
HMFOMUT	pH 7.0, 37 °C, 125 mM FAL, 8h	glucose 100 mM	97.0 %	This research

3.3. Chemobiological catalysis of D-xylose to FOL in LA:Bet-H_2O

In recent years, the built chemicbiocatalytic cascade process has attracted wide attention [8,11,26,37,47,56]. This combined process is important for effectually transforming inexpensive raw materials into valuable chemicals, which will reduce equipment investment and performance time [56–60]. Very recently, it is of great interest to manufacture furan-based chemicals from biomass or biobased derivatives in an efficient and sustainable reaction system [61,62]. In this research, a hybrid conversion through the sequential catalysis with LA:Bet and HMFOMUT cell was utilized to synthesize FOL from *D*-xylose. LA:Bet (10 wt%) and *D*-xylose (0.75 g) were uniformly mixed in 50 mL of water and reacted at 180 °C for 30 min to convert 81.6 mM FAL. The system's pH was further adjusted to 7.0, and the FAL concentration was diluted to 71.35 mM. Afterwards, HMFOMUT whole-cell biocatalysis with cosubstrate glucose (0.8 mol glucose/mol FAL) was carried out at 37 °C. The conversion rate of FAL was up to 100 %, and FAL could be converted into 68.4 mM FOL (95.8 % yield, 0.384 g FOL) and 3.0 mM FA within 24 h (Fig. 6a). The whole cells of HMFOMUT could effectually catalyze the conversion of *D*-xylose-derived FAL to FOL. Fig. 7 depicted the mass balance of *D*-xylose-to-FAL. 0.75 g *D*-xylose and 5 g LA:Bet were thoroughly blended in 50 mL water and added to a 100 mL autoclave. The

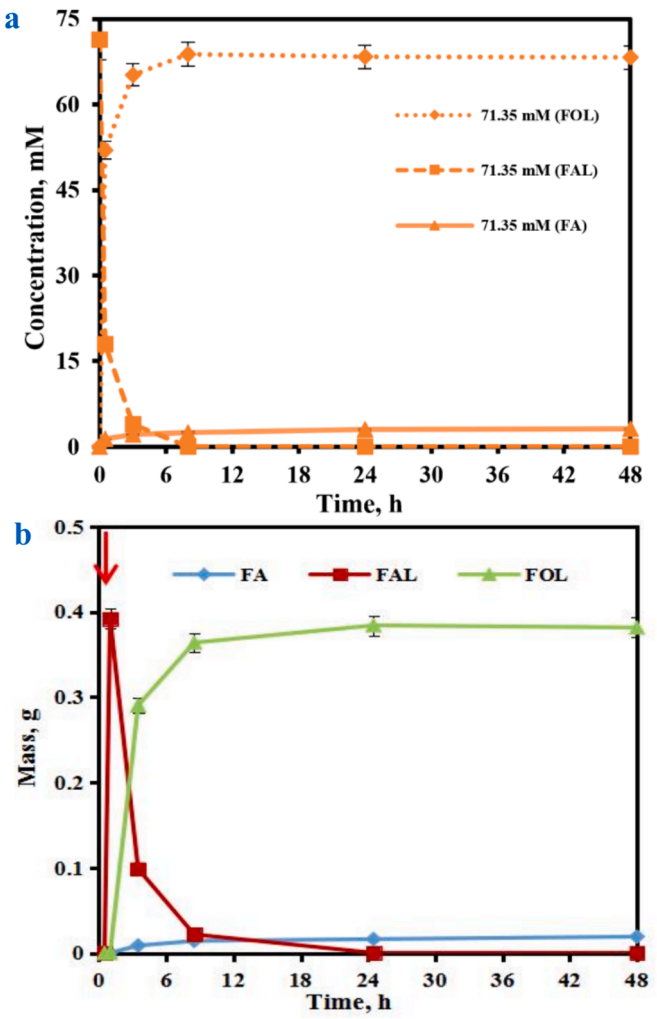


Fig. 6. Biotransformation of *D*-xylose-valorized FAL (81.60 mM and 71.35 mM) with HMFOMUT cells in LA:Bet-H_2O supplemented with co-substrate glucose (FAL-to-glucose molar ratio, 1:0.8) (a); Transformation of *D*-xylose into FOL in LA:Bet-water (b) [Arrow: pH adjustment, addition of HMFOMUT cells and co-substrate 0.8 mol glucose/mol FAL; Chemocatalysis of *D*-xylose at 180 °C in 30 min; Biocatalysis of *D*-xylose-valorized HMF at 37 °C and pH 7.0].

reaction at 180 °C for 30 min yielded a slurry containing 0.392 g FAL liquid. FAL solution derived from *D*-xylose was catalyzed by 2.859 g HMFOMUT wet cells with 0.588 g glucose as cosubstrate to form FOL (0.384 g) and FA (0.019 g) (Fig. 6b). A strategy for the synthesis of FOL from *D*-xylose by using LA:Bet and HMFOMUT as catalysts was effectually established.

In a concise summary, one tandem catalysis with DES LA:Bet (as chemocatalyst) and HMFOMUT cell (as biocatalyst) in the LA:Bet-H_2O system could effectually transform *D*-xylose into FOL in a high yield (molar yield 78.2 %, based on *D*-xylose), achieving chemobiological catalysis of bio-based chemicals into valuable furan-based coumpounds. The built chemobiological method could effectually convert *D*-xylose into FOL in the same reactor, avoiding the harsh performance conditions (high temperature, high pressure and polluted environment in the chemocatalytic process), frcrasing performance cost, and enhancing the xylose-to-FOL productivity, as well as reducing the investment of equipment [8]. DES, as a green catalytic solvent with simple preparation, might effectually avoid the input of excessive acidic catalysts and provide a new idea for the efficient use of biomass resources to synthesize high-value-added biobased compounds in a benign and sustainable system [42]. Accordingly, this research devised a one-pot chemobiocatalytic cascade strategy for the effective conversion of *D*-xylose to FOL via LA:Bet chemocatalysis and HMFOMUT cell biocatalysis, which holds significant promising for reducing the use of equipment, avoiding excessive water usage and efficiently promoting the biomass valorization.

4. Conclusion

The effect of DES LA:Bet-water on whole-cells catalytic FAL synthesis of FOL by HMFOMUT was testified. In the LA:Bet-H_2O (5:95, wt/wt) system, the whole cells of HMFOMUT showed the best catalytic performance and could effectually catalyze the synthesis of FOL from FAL at 50–125 mM when glucose (0.8 mol glucose/mol FAL) was used as a cosubstrate at pH 7.0 and 37 °C. The FOL yield reached 90.2 %-96.8 %. Afterwards, a one-pot tandem catalyzed dehydration of *D*-xylose for the synthesis of FOL using a chemo-biological method using LA:Bet and HMFOMUT cells as catalysts. *D*-Xylose (0.75 g/L) was converted to FAL in green DES LA:Bet (15 wt%) medium at 180 °C, and *D*-xylose-derived FAL was biologically converted to FOL in 95.8 % yield (i.e. 78.2 %, based on *D*-xylose). In this research, a one-pot tandem-catalyzed *D*-xylose synthesis of FOL with high efficiency was developed using LA:Bet and HMFOMUT as catalysts in LA:Bet-H_2O.

CRediT authorship contribution statement

Shunli Zhang: Writing – original draft, Validation, Software, Resources, Methodology, Investigation, Data curation, Conceptualization. **Yu-Cai He:** Writing – review & editing, Supervision. **Cuiluan Ma:** Writing – review & editing, Supervision.

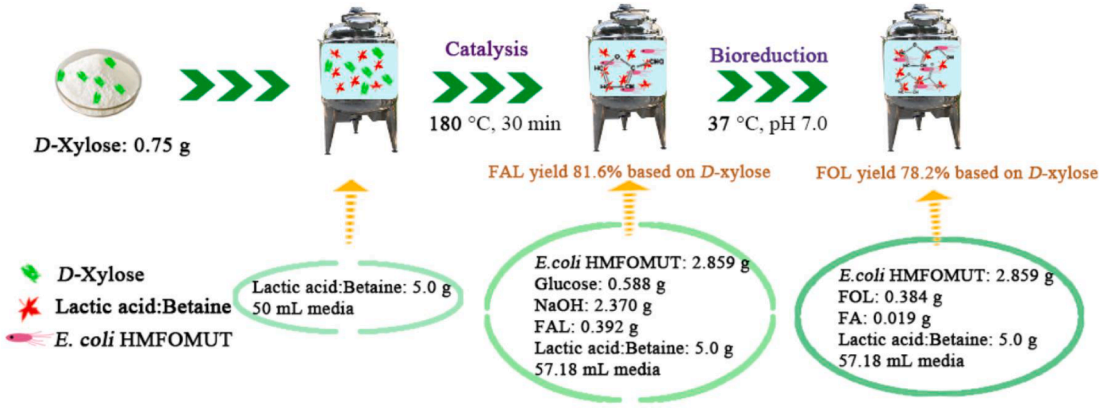


Fig. 7. Material balance diagram.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

No data was used for the research described in the article.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.molliq.2024.125576.

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